Dang Tran

tranquangthiendang2003@gmail.com | linkedin.com/in/dangtran278 | github.com/dangtran278

Education

University of Massachusetts Amherst — Amherst, MA

Master of Science in Computer Science, Specialized in Machine Learning — *Sep. 2025 – May 2026*

- **GPA**: 3.84/4.00 • **Awards**: Bay State Scholar

Bachelor of Science in Computer Science, Minor in Mathematics — *Sep. 2021 – May 2025*

- **GPA**: 3.94/4.00 • **Awards**: Dean's List, Chancellor's Award, HackHolyoke Best Health Hack

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, R, SQL, HTML/CSS

Machine Learning: Deep Learning, Computer Vision, Natural Language Processing (NLP), Information Retrieval, Large Language Models (LLMs), Generative AI, Multimodal Learning, Vision-Language Models (VLMs), Agentic AI, Fine-tuning, Retrieval-Augmented Generation (RAG), Prompt Engineering, Inference Optimization, Model Deployment

Libraries/Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, vLLM, LangChain, Scikit-Learn, FAISS, OpenCV, NumPy, Pandas, Matplotlib, FastAPI, Flask, REST API, Airflow, pytest

Cloud/Tools/Databases: Git, Docker, AWS (ECS, ECR), GitHub Actions, CI/CD, Prometheus, Slurm (GPU cluster), Qdrant (vector database), PostgreSQL

Experience

Adobe — Feb. 2026 – May 2026

Research Extern | Co-first author, MultAttnAttrib (Under Review, EACL 2027) — *Remote*

- **Outperformed GPT-5.4 on visual grounding** with a training-free algorithm that reads attention heads of Qwen3-VL-30B, 7.3× faster than prompting it
- Automated generation of a 2,285-pair attribution benchmark across 6 domains, running vLLM inference across distributed Slurm nodes
- Engineered a multi-stage RAG pipeline over Cohere text-image retrieval and cross-modal entity grounding, gated by rubric-based LLM-as-judge

UMass BioNLP Lab — Sep. 2025 – Dec. 2025

Graduate Researcher — *Amherst, MA*

- Curated a medical Visual Question Answering (VQA) benchmark of 500+ samples from 5 datasets, establishing baselines for 6 VLMs
- **Boosted zero-shot clinical accuracy by 6 points**, refining agent's tool selection: BiomedCLIP (classify), Grounding DINO (localize), MedSAM (segment)
- Lowered evaluation latency by ~90%, serving all 6 VLMs through vLLM batched inference

CB/I Digital — June 2025 – Aug. 2025

Data Scientist Intern — *Remote*

- **Reduced new-product forecasting error by 12% wMAPE** across 6 brands, matching similar products with multimodal embeddings and ensemble averaging
- Built an anomaly detection pipeline for 600+ SKUs, flagging demand shifts with Z-score and residual analysis

FPT Information System — June 2023 – Aug. 2023

Machine Learning Engineer Intern — *Hanoi, Vietnam*

- **Cut retrieval time by ~70%** with a RAG pipeline of bi-encoder retrieval and cross-encoder reranking
- Achieved 85% recall and 81% precision with FAISS + BM25 search and a PyTorch fine-tuned BERT reranker

MobiFone — June 2022 – Aug. 2022

Computer Vision Engineer Intern — *Cantho, Vietnam*

- Improved legal document OCR speed by 20% with U-Net layout segmentation and fuzzy field matching
- **Deployed the OCR pipeline as a Flask REST API**, automating JSON extraction from 200+ legal documents

Projects

Ctrl F-ilm (Search Engine for Video) | *PyTorch, Qdrant, AWS, Docker, CI/CD, FastAPI* — Aug. 2026 – Present

- **Launched a video search engine** that returns the matching clip and its timestamps for a plain-text query
- Reached 38% top-1 and 76% top-10 zero-shot retrieval on 1,019 YouTube videos from ActivityNet
- Automated commit-to-production deploys on AWS ECS behind a 3-stage CI/CD gate